_/_/_

Discrete Event Simulation

General Principles

The basic building blocks of all discrete-event simulation models : entities and attributes, activities and events.

A system is modelled in terms of
- Its state at each point in time.
- The entities that pass through the system and the entities that represent system resources
- The activities and events that cause system state to change.

Discrete event models are appropriate for those systems for which changes in system state occur only at discrete points in time.

- Deals with dynamic, stochastic systems (ie. involving time and containing random elements) which change in a discrete manner

Model : An abstract representation of a system usually containing structural, logical or mathematical relationships which describe a system in terms of state, entities and their attributes, sets, processes, events, activities and delays.

List : A collection of (permanently or temporarily) associated entities, ordered in some logical fashion

Event

* **Event Notice** → a record of an event to occur at the current or some future time, along with any associated data necessary to execute the event, at a minimum, the record includes the event type and the event time.

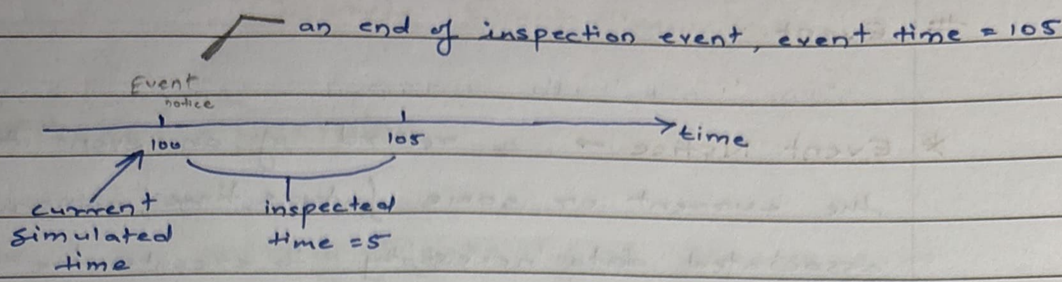
* **Event List** → A list of event notices for future events, ordered by time of occurrence also known as future event list (FEL)

Delay → A duration of time of unspecified indefinite length, which is not known until it ends

* Clock → A variable representing simulated time, called CLOCK in the examples to follow

An activity typically represents a service time, an inter arrival time or any other processing time whose duration has been characterized and defined by the modeler

An activitiy's duration may be specified in a no. of ways:
- Deterministic - for a example, always exactly 5 mins.
- Statistical - for example - as a random draw amongst 2, 5, 7 with equal probabilities.
- A function depending on system variables and/or entity attributes → for example, loading time for an iron ore ship as a function of the ship's allowed cargo weight and loading rate in tons per hour



The duration of an activity is computable from its specification at the instant it begins.

To keep track of activities and their expected completion time, at ~~activity~~ the simulated instant that an activity duration begins, an event notice is created having an event time equal to the activity's completion time

* Delay's Duration
  - is not specified by the modeler ahead of time, but rather determined by system conditions
  - A delay's duration is measured and is one of the desired outputs of a model run.

eg. A customer's delay in a waiting line may be dependent on the number ~~of~~ and duration of service of other customers ahead in line as well as the availability of servers and equipment.

	Delay	Activity
What so called	a conditional wait	an unconditional wait
A completion	a secondary event ~~wait~~	a primary event
A management	by placing an event notice on the FEL	by placing the associated entity on another list, not the FEL, perhaps representing a waiting line

Aju nice

→ System state, entity attributes and the number of active entities, the contents of sets and the activities and delays currently in progress are all functions of time and are constantly changing over time.

→ Time itself is represented by a variable called CLOCK

→ A discrete event simulation: the modelling over time for a system all of whose state changes occur at discrete points in time - those points when an event occurs.

→ It proceeds by producing a sequence of system snapshots which represent the evolution of the system through time

Event Scheduling / Time Advanced Algorithm

↳ The mechanism for advancing simulation time and guaranteeing that all events occur in correct chronological order is based on the future event list

* Future Event List

↳ to contain all event notices for events that have been scheduled to occur at a future time.

↳ to be ordered by event time, meaning that the events are arranged chronologically; that is, the event times satisfy : $t < t_1 \leq t_2 \leq \wedge \leq t_n$

current value of simulated time

Imminent event

↳ Scheduling a future event means that at the instant an activity begins, its distribution and the end-activity event, together with its event time is placed on the future event list.

Old system at time t

Clock	System State	..	Future Event List	...
t	(5,1,6)		(3, t_1), (1, t_1), (1, t_3)..	

Event Scheduling Algorithm

Step 1: Remove the event notice for the imminent event from FEL

Step 2: Advance CLOCK to imminent event time

Step 3: Execute èmminent event, update system state, change entity attribute's and set membership.

Step 4: Generate future events and place their event notices on FEL ranked by event time

Step 5: Update cumulative statistics and counter

CLOCK	System State	...	future Event List	...
t_1	(5,1,5)		(1, t_2), (4, t^*) (1, t_3)..	

List processing : management of list

↳ the removal of imminent event

↳ as the imminent event is usually at the top of the list, its removal is as efficient as possible

↳ The addition of a new event to the list, and occassionally removal of some event

↳ addition of a new event requires a search of the list.

→ The efficiency of this search depends on the logical organization of the list and on how the search is conducted.

→ The system snapshots at time 0 is defined by the initial conditions and the generation of the so-called exogeneous events.

* An exogeneous event → happening outside system

How are future events generated?

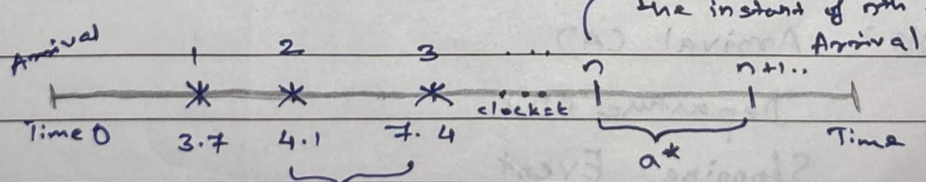
↳ to generate an arrival to a queueing system

↳ by a service-completion event in a queueing simulation

↳ to generate runtimes and downtimes for a machine subject to breakdowns

* To generate an arrival to a queueing system

at simulated time t, assumed to be the instant of nth arrival



Between successive arrival events other type of events may occur causing system state to change

__/__/__

By a service completion event in a queueing simulation
↳ A new service time s^*, will be generated for the next customer
↳ when one customer completes service, at current time CLOCK = t
↳ If the next customer is present

→ The next service-completion event will be scheduled to occur at future time $t^* = t + s^*$ by placing onto the FEL a new event notice ~~notice~~ of type service completion

→ A service completion event will be generated and scheduled at the time of an arrival event, provided that, upon arrival, there is at least one idle server in the server group.

Single Channel Queue

→ System State → LQ(t), LS(t)
↳ LQ(t) = no. of customers in the waiting line
↳ LS(t) = no. of being served (0 or 1) at time t

→ Entities → the server and customer are not explicitly modeled, except in terms of the state variables ~~with~~

→ Events : Arrival (A)
Departure (D)
Stopping Event

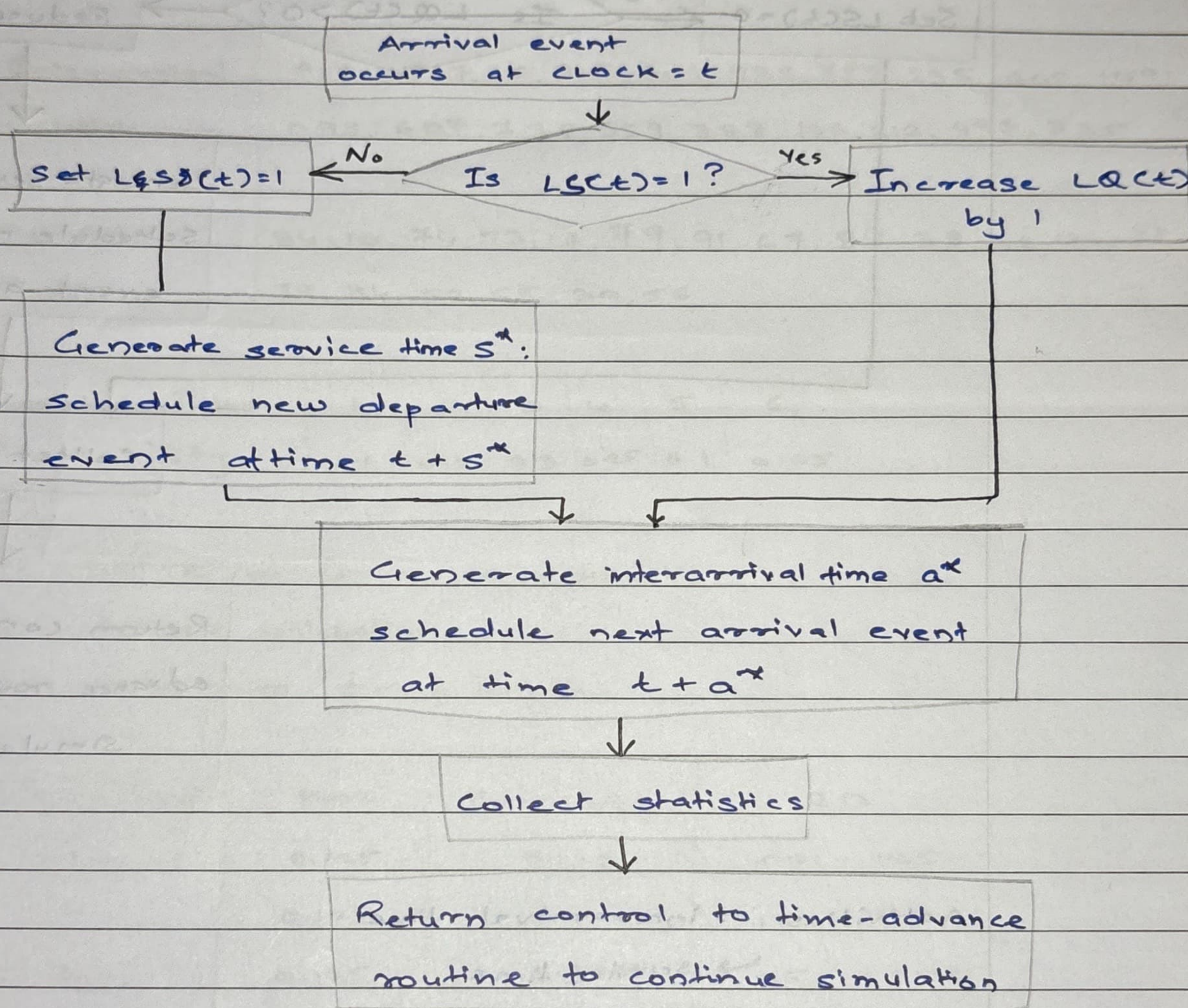
Event notices (Event type, Event notice)

↳ (A, t) → Arrival at future time t

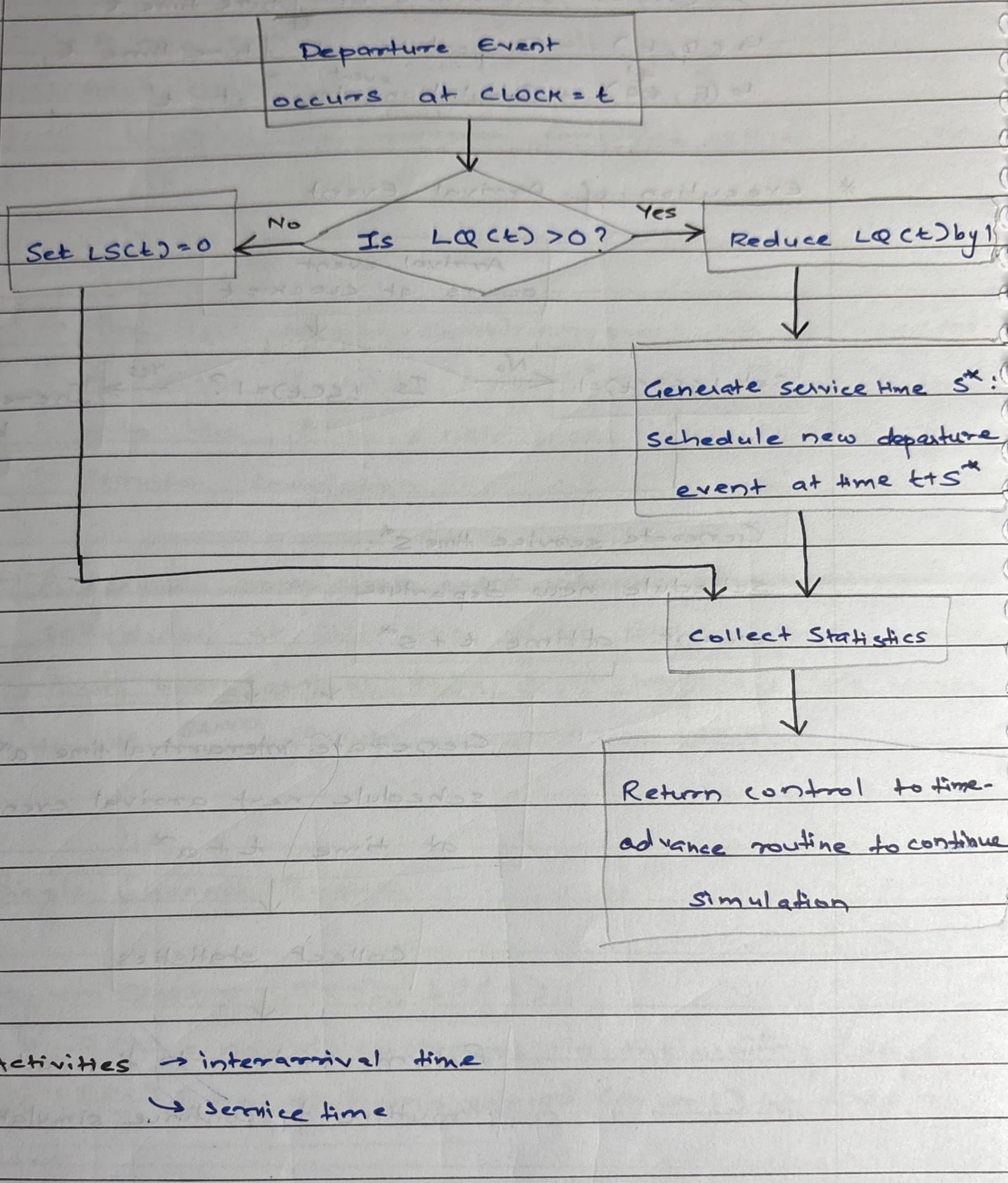
↳ (D, t) → Departure at future time t

↳ (E, t') → Simulation event stop at t'

* Execution of Arrival Event



* Execution of the departure event



Activities → interarrival time
→ service time

__/__/__

Q A small grocery store has only 1 checkout counter. Customers arrive at this checkout counter at random from 1 to 8 minutes apart. Each possible value of interarrival time has the same prob. of occurence. The service time varies from 1 to 6 minutes Simulate table for arrival of 20 customers

Interarrival Time	913, 727, 15, 948, 309, 922, 753, 235, 302, 109, 093, 607, 738, 359, 888, 106, 212, 493, 535
Service Time	84, 10, 74, 53, 17, 79, 91, 67, 89, 38, 32, 94, 79, 0 79, 84, 52, 55, 30, 56

Service Time	1	2	3	4	5	6
Probability	0.1	0.2	0.3	0.25	0.1	0.05

Determine performance measures
(i) Server utilization
(ii) maximum queue length

Inter arrival	Prob	Cumulative	RD
1	0.125	0.125	001 - 125
2	0.125	0.250	126 - 250
3	0.125	0.375	251 - 375
4	0.125	0.500	376 - 500
5	0.125	0.675	501 - 675
6	0.125	0.750	676 - 750
7	0.125	0.875	751 - 875
8	0.125	1.000	876 - 000

_/_/_

Service Time Probability Distribution

Service Time	Probability	Cumulative Probability	R D
1	0.1	0.1	01-10
2	0.2	0.3	11-30
3	0.3	0.6	31-60
4	0.25	0.85	61-85
5	0.1	0.95	86-95
6	0.05	1.00	96-00

RD	Interarrival	RD Service Time	ST
913	~~8~~	84	~~4~~
727	~~6~~	10	~~1~~
15	~~1~~	14	~~4~~
948	~~8~~	53	~~3~~
309	~~3~~	17	~~2~~
922	~~8~~	79	~~4~~ 4
753	~~7~~	91	~~5~~
235	~~3~~	67	~~4~~

Assume simulation is ending at time = 60

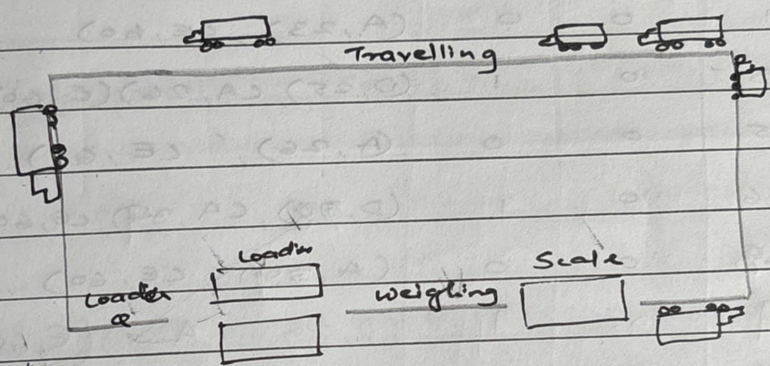
302	~~3~~	89	~~5~~
109	~~1~~	38	~~3~~
093	~~1~~	32	~~3~~
607	~~8~~	94	5
738	~~6~~	79	4
359	3	05	1
888	8	79	4
106	1	84	4

D= Departure, A= Arrival

	Clock	System State LQ(t)	LS(t)	FEL	Statistics Busy State (Depart time dif) (old clock - new clock)	Max Queue
A	0	0	1	(D,4), (A,8), (E,60)	0	0
D	4	0	0	(A,8), (E,60)	4	0
A	8	0	1	(D,9), (A,14), (E,60)	4	0
D	9	0	0	(A,14), (E,60)	5	0
A	14	0	1	(A,15), (D,18), (E,60)	5	0
A	15	1	1	(D,18), (A,23), (E,60)	6	1
D	18	0	1	(D,21) (A,23) (E,60)	9	1
D	21	0	0	(A,23), (E,60)	12	1
A	23	0	1	(D,25), (A,26) (E,60)	12?	1
D	25	0	0	(A,26), (E,60)	12+2=14?	1
A	26	0	1	(D,28), (A,29), (E,60)	14	1
D	28	0	0	(A,29), (E,60)	16	1
A	29	0	1	(D,33), (A,37) (E,60)	16	1
D	33	0	0	(A,37) (E,60)	20	1
A	37	0	1	(D,42), (A,44) (E,60)	20	1
D	42	0	0	(A,44), (E,60)	25	1
A	44	0	1	(A,46), (D,48) (E,60)	25	1
A	46	1	1	(D,48) (A,49) (E,60)	27	2
D	48	0	1	(A,49) (D,53) (E,60)	29	2
A	49	1	1	(A,50) (D,53) (E,60)	29	2
A	50	2	1	(A,51), (D,53), (E,60)	30	3
A	51	3	1	(D,53) (A,56), (E,60)	31	4
D	53	2	1	(D,56) (A,56), (E,60)	33	4
A	56	3	1	(D,59) (E,60) (A,62)	36	5
D	56	2	1	(D,59) (E,60), (A,62)	36	5
D	59	1	1	(E,60) (A,62)	39	5
E	60	1	1	(E,60) (A,62)	39	5

__/__/__

Q 6 dump trucks are used to haul coal from the entrance of a small mine to the rail road. Each truck is loaded by one of the 2 loaders. After loading, a truck immediately moves to the scale to be weighed as soon as possible. Both are the loaders and scale have FIFO waiting lines for trucks. Travel time from loader to a scale is considered negligible. After being weighed, a truck begins a travel time and then afterwards returns to loader queue. The distribution of loading time, weighing time and travel time are as follows.



Loading Time	Prob	Cumu	RD
5	0.3	0.3	01-03
10	0.5	0.8	04-08
15	0.2	1.0	09-00

Weighing Time	Prob	Cumu	RD
12	0.7	0.7	01-07
16	0.3	1.0	08-00

_/_/_

Travel Time	Prob	Cumul	RD
40	0.4	0.4	01-04
60	0.3	0.7	05-07
80	0.2	0.9	08-09
100	0.1	1.0	00

RD for Loading	4, 1, 2, 5, 9, 6, 7
RD for weighing	1, 2, 3, 8, 5, 9
RD for traveling	5, 0, 1, 2, 8

Assume that 5 trucks are at the loaders and one is at the scale at time 0. Simulate for 20 minutes

Loading		Weighing		Traveling	
4	10	1	12	5	60
1	5	2	12	0	100
2	5	3	12	1	40
5	10	8	16	2	40
9	15	5	12	8	80
6	10	9	16		
7	10				

__/__/__

CLOCK	System State				List				
	LQ(t)	L(t)	WQ(t)	W(t)	LoadQ	WeighQ	FEL	BL	BS
0	3	2	0	1	DT4, DT5 DT6	-	(EW, 12, DT1) (EL, 10, DT2) (EL, 5, DT3)	0	0
5	2	2	1	1	DT5, DT6	DT3	(EW, 12, DT1) (EL, 10, DT2) (EL, 10, DT4)	10	5
10	1	2	2	1	DT6	DT3, DT2	(EW, 12, DT1) (EL, 10, DT4) (EL, 10+10, DT4)	10+10	10
10	0	2	3	1	-	DT3, DT2 DT4	(EW, 12, DT1), (EL, 20, DT5) (EL, 10+15, DT6)	20	10
12	0	2	2	1	~~DT3~~, —	DT2, DT4	(EL, 20, DT5) (EW, 12+12, DT3) (EL, 25, DT6) (ALQ, 12+60, DT1)		

__/__/__

Probability and Statistics in Simulation

- Probability theory is the study of random phenomenon, represented in terms of mathematical description by constructing ideal probalistic model of real world simulation.
- Ideal - consists of all possible outcomes and corresponding pro

Role of Probability theory
- is to analyze the behavior of system as per given distribution and corresponding probability

Statistical Theory → is helpful in choosing these probability and validating the model assumption

Why do we need probability and statistics in simulation?
- to validate the simulation model
- to determine/choose input probability distributions
- to perform statistically analyze output data/results
- to design correct/efficient simulation experiments

Experiment → a process which could result in several different outcomes

Sample Space → set of possible outcomes of a given expeiment

Example : Rolling a dice → {1, 2, 3, 4, 5, 6}

Random Variable → a function that assigns a real no. to each point in a sample space

Discrete Random ~~Variable~~ Variable → rv is 1 for which the no. of possible values is finite or countably finite

For each possible value x_i, for discrete random variable X, there is $P(X = x_i) = p(x_i)$

$p(x_i) \rightarrow$ probability mass function

$p(x_i) \geq 0$ for all i

$\Sigma p(x_i) = 1$

Cumulative Distribution Function (cdf), $F(x)$

$F(x) = P(X <= x)$

$F(x) \sum_{x_i \leq x} p(x_i)$

Properties :

$0 \leq F \leq 1$

$\lim_{x \to -\infty} F(x) = 0$, $\lim_{x \to \infty} F(x) = 1$

f is non decresing

$P(a < x \leq b) = F(b) - F(a)$ for all $a < b$

Variance, $V(X) = E(X^2) - [E(X)]^2$

$E(X) = \Sigma p(x_i) \cdot x_i$ → mean

Q What is the variance of the following group of exam scores : {75, 90, 40, 95, 80}

→ $E(X) = \sum x_i \cdot p(x_i)$

$V(X) = E(X^2) - [E(X)]^2$

$E(X) = \frac{1}{5} \times 75 + \frac{1}{5} \times 90 + \frac{1}{5} \times 40 + \frac{1}{5} \times 95 + \frac{1}{5} \times 80$

$= 15 + 18 + 8 + 19 + 16$

$= 76$

$E(X^2) = \frac{1}{5} \cdot 5625 + \frac{1}{5} \cdot 8100 + \frac{1}{5} \cdot 1600 + \frac{1}{5} \cdot 9025 + \frac{1}{5}\ 6400$

$= 6150$

$V(X) = 6150 - 76^2 = 6150 - 5776 = 374$

Discrete Distributions of Interest

* Bernoulli: Tracks and Bernoulli Distribution

- consider an experiment with the following properties
  - n independent trials are performed
  - each trial has 2 possible results - success or failure
  - probability of success, p and failure, $q = 1-p$ is constant from trial to trials
  - for random variable, X, $X=1$ for a success and $X=0$ for a failure

$E(X) = pn$

$V(X) = pqn$

A single Bernoulli trial is not that interesting

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0, 1, K, n \\ 0 & \text{otherwise} \end{cases}$$

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Generally speaking binomial distributions can be used to determine the probability of a given number of defective items in a batch, or the probability of a given number of people having a certain characteristic

ex: The trait of having a klinkled flooje occurs on average in 10% of kreptoplomians. Given a ~~people~~ group of 20 kreptoplomians. What is the probability that 3 of them have klinkled floojes?

$$p(3) = \frac{20!}{3!17!} \times p^3 q^{17} = 0.190119$$

* Geometric Distribution

$$p(x) = \begin{cases} q^{x-1} p, & x = 1, 2, \ldots \\ 0, & \text{otherwise} \end{cases}$$

$$F(x) = 1 - q^x$$

$$E(x) = 1/p$$

$$V(x) = \frac{q}{p^2}$$

* Poisson Distribution

$$p(x) = \begin{cases} \frac{e^{-\alpha} \alpha^x}{x!} ; & x = 0, 1 \ldots \\ 0 & \text{otherwise} \end{cases}$$

$$E(x) = \alpha = V(x)$$

$$F(x) = \sum_{i=0}^{x} \frac{e^{-\alpha} \alpha^i}{i!}$$

__/__/__

Q A computer repair person is "beeped" each time there is a call of service. The number of beeps per hour is known to occur in accordance with Poisson distribution with a mean of $\alpha = 2/hr$. The probability of 3 beeps in the next hour with $x = 3$:

$$\rightarrow p(3) = \frac{e^{-2} 2^3}{3!} = \frac{(0.135)(8)}{6} = 0.18$$

* Uniform Distribution

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

$$\text{CDF}, \quad F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x < b \\ 1 & x \geq b \end{cases}$$

$$E(X) = \frac{a+b}{2}$$

$$V(X) = \frac{(b-a)^2}{12}$$

* Exponential Distribution

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$E(X) = \frac{1}{\lambda}, \quad V(X) = \frac{1}{\lambda^2}$$

$$F(x) = \begin{cases} 0 & , x < 0 \\ \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x} & , x \geq 0 \end{cases}$$

__/__/__

Q Assume that hard drive manufactured by Herb's Hard Drives have a mean life time of 4 years

a) What is the probability that one of Herb's Hard Drives will fail within the first 2 years?

b) What is the probability that one of Herb's Hard Drives will last at least 8 yrs?

→ a) $P(x \leq 2) = F(2) = 1 - e^{-1/4 \cdot 2} = 1 - e^{-1/2} = 1 - 0.6065$

$= 0.3934$

b) $P(x > 8) = 1 - P(x \leq 8) = 1 - (1 - e^{-1/4 \cdot 8})$

$= 0.1353$

* Normal Distribution

$\lim_{x \to -\infty} f(x) = 0 = \lim_{x \to \infty} f(x)$

$f(\mu + x) = f(\mu - x)$ → $\max(f(x)) = f(\mu)$

For $\phi(z)$ → normal distribution,

mean $= \mu = 0$, variance $(\sigma^2) = 1$

$= N(0,1)$ → standard normal distribution

Notation : $Z \sim N(0,1)$

$X \sim N(\mu, \sigma^2)$

$z = \dfrac{x - \mu}{\sigma}$

$f(x) = \phi\left(\dfrac{x-\mu}{\sigma}\right)$ → $\phi(z) = \dfrac{1}{\sqrt{2\pi}} e^{-z^2/2}$

_/_/_

* Triangular Distribution

PDF, $f(x) = \begin{cases} \dfrac{2(x-a)}{(b-a)(c-a)} & , a \leq x \leq b \\ \dfrac{2(c-x)}{(c-b)(c-a)} & , b \leq x \leq c \\ 0 & , \text{elsewhere} \end{cases}$

CDF, $F(x) = \begin{cases} 0 & x \leq a \\ \dfrac{(x-a)^2}{(b-a)(c-a)} & a < x \leq b \\ 1 - \dfrac{(c-x)^2}{(c-b)(c-a)} & b < x \leq c \\ 1 & x > c \end{cases}$

$E(x) = \dfrac{a+b+c}{3}$, Mode $= b$

$V(x) = \dfrac{a^2+b^2+c^2-ac-bc-ab}{18}$

Q An industrial chemical that will retard the spread of fire in paint has been developed. The local sales representative has determined from the past experience that 48% of the sales calls will result in an order will ~~[illegible]~~ result in an order

a) What is the probability that the first order will come on the 4th sales call of the day?

b) If 8 sales calls are made in a day. What is the probability of recieving exactly 6 orders?

c) If 4 sales calls are made before lunch, what is the prob that one or less results in an order?

_/_/_

→ a) Geometric ~~Degree~~ Distribution,

$p = 0.48$, $q = 0.52$

$p(4) = (0.52)^{4-1} (0.48) = 0.067491$

b) Binomial Distribution

$n = 8$, $q = 0.52$, $p = 0.48$

$p(6) = {}^8C_6\, p^6 q^2 = 0.0926$

c) Binomial Distribution

$n = 4$, $p = 0.48$, $q = 0.52$

$p(x \le 1) = p(x = 1) + p(x = 0)$

$= {}^4C_1\, p^1 q^3 + {}^4C_0\, p^0 q^4$

$= 0.3431$

Q Joe Coledge is the third string quarterback for university of Lower Alatoon. The prob tha Joe gets into a game is 0.40.

a) What is the prob that the first game enters is the 4th game of the season

b) What is the prob that Joe plays in no more than 2 of the 1st 5 games.

→ a) Geometric

$p = 0.40$, $q = 0.60$

$p(4) = (0.6)^{4-1} (0.4) = 0.0864$

b) Binomial

$n = 5$, $p = 0.4$, $q = 0.6$

$p(x \le 2) = p(x = 0) + p(x = 1) + p(x = 2)$

$= {}^5C_0\, 0.4^0\, 0.6^5 + {}^5C_1\, 0.4 \times 0.6^4 + {}^5C_2\, 0.4^2 \times 0.6^3$

$= 0.68256$

_/_/_

Q The number of hurricanes hiting the coast of florida annualing has a poisson dist^n with a mean of 0.8.

a) What is the prob that more than 2 hurricanes will hit the florida coast in a year?

b) What is the prob that exactly 1 hurricane will hit the coast of florida in a year?

→ a) $p(x>2) = 1-(p(x=2)+p(x=1)+p(x=0))$

$$= 1 - \left[\frac{e^{-0.8} \times 0.8^2}{2!} + \frac{e^{-0.8} \times 0.8^1}{2!} + \frac{e^{-0.8} \times 1}{1} \right]$$

$= 0.04742259$

b) $p(x=1) = \frac{e^{-0.8} \times 0.8^1}{1} = 0.3595$

Q Arrivals at a banks tellers cage is poisson distri at a rate of 1.2/min.

a) What is the prob of 0 arrivals in the next min?

b) What is the prob of 0 arrivals in the next 2 mins?

→ a) $P(x=0) = \frac{e^{-m}\, m^x}{x!} = \frac{e^{-1.2} \times 1.2^0}{0!} = 0.301194$

b) $\alpha = 2 \times 1.2 = 2.4$

$P(x=0) = \frac{e^{-2.4} \times 2.4^0}{0!} = 0.0907179$

Q A production process manufactures alternators for outboard engines used in recreational boating. On the average, 1% of the alternators will not perform upto the reqd. standards when tested at engine assembly plank. When a ship next of 100 alternators is recieved at the plank, they are tested ~~of~~ and if more that 2 are non-confirming, the shipment is returned to the alternator manufacturer. What

__/__/__

is the probability of returning shipment?

→ $p = 0.01$, $q = 0.99$, $n = 100$

$x > 2$

$$p(x>2) = 1-(p(x=0)+p(x=1)+p(x=2))$$

$$= 1-\left[{}^{100}C_0 (0.01)^0 (0.99)^{100} + {}^{100}C_1 (0.01)^1 (0.99)^{99} + {}^{100}C_2 (0.01)^2 (0.99)^{98}\right]$$

$= 0.0794$

Q A passenger arrives at a railway station randomly which is uniformly distributed b/w 10 am and 10:30 am. Find the prob that the passenger has to wait for the train for

a) Less than 6 min

b) more than 10 min

The trains arrive for every 15 min beginning at 5 am

→ a) 10:9 to 10:15, 10:24 to 10:30

$$p(w<6) = \frac{6-0}{15-0} = \frac{2}{5} = 0.4$$

$$b)\ p(w>10) = \frac{15-10}{15} = \frac{5}{15} = 0.33$$

___/___/___

Q The components of a system whose time to failure is exponentialy distributed with failure rate $\lambda = 1/6$, if 6 such compoments are installed at different systems, what is the probability that 2 are still working at end of 9 years

→ $\lambda = \frac{1}{6}$, $t = 9$

P (1 comp after 6 yrs)

Q The time to pass through a queue to begin self service at a cafetaria has been found to be normally distributed given by $N \sim (15, 9)$. Determine prob that arriving cust waits between 14 and 17 mins

→ $\mu = 15$, $\sigma^2 = 9$, $\sigma = 3$

For $x = 14$, $z_1 = \frac{14 - 15}{3} = \frac{-1}{3} = -0.33$

for $x = 17$, $z_2 = \frac{17 - 15}{3} = \frac{2}{3} = 0.66$

$P(-0.33 < z < 0.67) = P(z < 0.67) - P(z < -0.33)$

$= 0.7486 - 0.3707 = 0.3780$

Q The central processing requirement for a program has triangular distribution where $a = 0.05$, $b = 1.1$, $c = 6.5$. Determine the probability of CPU requirement for a progress of 2.5 s. → $2.5 > 1.1$ (b)

→ $f(x) = \frac{2(c - x)}{(c - b)(c - a)}$ → $f(2.5) = \frac{2(6.5 - 2.5)}{(6.5 - 0.05)(6.5 - 1.1)} = \cancel{8}$

$= 0.229687$

* Poisson Arrival Process
  - arrivals occur one at a time
  - number of arrivals in a given time depends only on length of that period and not on starting point
  - no. of arrivals in a given time period does not affect no. of ——— in a subsequent period
  - 〃 ——— are independent of each other

We can alter it to include time:

$$P(N(t) = n) = \begin{cases} \dfrac{e^{-\lambda t} (\lambda t)^n}{n!} & , n = 0, 1, \ldots \\ 0 & , \text{otherwise} \end{cases}$$

$V = E = \alpha = \lambda t$

Properties:
  - Random Splitting
  - Splitting
  - Pooling

When a random variable has no known distribution, it is called as a Emphenical Distribution

Random Variate Generation
  - essential in DE Simulation
  - random observations of a random variable that form a desired probability distribution are called as Random Variates

$$\boxed{\text{Uniform Random Number Generation}} \longrightarrow \boxed{\text{Convert to desired Distribution}} \longrightarrow \boxed{\text{RD}}$$

Algorithm :

↳ Get the CDF of desired random variable x

↳ Generate $R \sim U\{0,1\}$ and set $F(x) = R$ on range of x

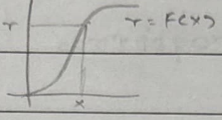
↳ calculate F^{-1} by solving eq^n $F(x) = R$ in terms of $R \Rightarrow x = F^{-1}(R)$

↳ Generate uniform random no.s $R_1, R_2, R_3 \ldots$ from $U[0,1]$ and use $X_i = F^{-1}(R_i)$ to generate variates

Inverse Transform

cdf, $r = F(x)$

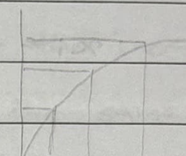
$x = F^{-1}(r)$



* Exponential Distribution

$R = F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$

↳ $X_i = F^{-1}(R_i) = \frac{-1}{\lambda} \ln(1 - R_i)$



Q A system has a component whose time to failure is exponentially distributed with a failure rate of 1/6. Generate 2 random failure times from distribution. Use $R_1 = 0.38$, $R_2 = 0.45$

→ $X_1 = \frac{-1}{(1/6)} \ln(1 - 0.38) = 2.868$

$X_2 = \frac{-1}{(1/6)} \ln(1 - 0.45) = 3.587$

* Uniform Distribution

$f(x) = \frac{1}{b-a}$, $F(x) = \frac{x-a}{b-a}$ if $a \leq x \leq b$

$F(x) = r$

$x - a = r(b-a)$

$x_i = r_i(b-a) + a = a + r_i(b-a)$

__/__/__

Q The time reqd for a person to travel from home to college is uniformly distributed over the interval 30 to 35 mins. ~~Calculate~~ Generate 2 random ~~variables~~ travel times for this distribu^n

$R_1 = 0.5023$, $R_2 = 0.2916$

$\rightarrow b = 35, a = 30$

$X_1 = 30 + 0.5023(5) = 32.5115$

$X_2 = 30 + 0.2916(5) = 31.458$

* Triangular

$$x_i = \sqrt{cR_i} \quad \text{if } 0 < R \leq \frac{1}{c}$$

$$x_i = c - \sqrt{c(c-1)(1-R_i)} \quad \text{if } \frac{1}{c} < R_i \leq 1$$

Q A digital sensor is used to determine the quality of computer chips. The QC dept rejects chips those which face which is triangular distributed with $a=0$, $b=1$, $c=3$. Generate 2 random rejected samples from distribution.

Use $R_1 = 0.9536$, $R_2 = 0.2941$

$\rightarrow \frac{1}{c} = \frac{1}{3} = 0.33$

$R_1 > \frac{1}{c}$

$0 < R_2 \leq 1/c$

$X_1 = 2.472$

$X_2 = 0.93931$

~~$X_2 = \sqrt{3 \times 0.2941}$~~

Empirical Continuous Distribution

↳ when theoretical distribution is not ~~acceptable~~ applicable

↳ to collect empirical data

↳ resample the observed data

↳ interpolate b/w observed data points to fill in the gaps

↳ Use table look up method for generating random variates.

↳ A frequency distribution table is generated showing the interval and its corresponding probability, cumulative prob, slope

↳ Since all values between the tabulated ~~values~~ observations, the desired random variate is taken to be output value at the lower tabulated observation + the increment that divides the output interval in the same proportion that the input divides the input interval

For small sample set (size n):

↳ arrange in ascending order

↳ assign probability 1/n to each intervals

$$X = \hat{F}^{-1}(R) = x_{i-1} + a_i \left(R - \frac{(i-1)}{n} \right)$$

$$a_i = \frac{x_i - x_{i-1}}{1/n}$$

Grouped data:

$$a_i = \frac{x_i - x_{i-1}}{c_i - c_{i-1}}$$

$$X = F(x) = x_{i-1} + a_i (R - c_{i-1}) \quad \text{where} \ \frac{(i-1)}{n} < R \leq \frac{1}{n}$$

_/_/_

Q 5 observations of a fire station response time to the incoming alarms have been collected. The response times are 1.60, 0.55, 1.12, 2.20, 1.94. Setup a table for the generation of response time by table lookup method and generate 2 values of response time using uniform random no.s $R_1 = 0.5426$, $R_2 = 0.1524$

→

	Interval $x_{i-1} < x < x_i$	Prob	Cum Prob	Slope $= (x_i - x_{i-1}) / 1/n$
1	$0 < x \leq 0.55$	0.2	0.2	2.75
2	$0.55 < x < 1.12$	0.2	0.4	2.85
3	$1.12 < x \leq 1.60$	0.2	0.6	2.4
4	$1.60 < x < 1.94$	0.2	0.8	1.7
5	$1.94 < x < 2.20$	0.2	1.0	1.3

$$X_i = x_{i-1} + a_i (R - (i-1)/n)$$

$R_1 = 0.5426$, $R_2 = 0.1524$

(look in cum prob)

$$X_1 = x_2 + 2.4 \left(0.5426 - \frac{(3-1)}{5}\right)$$

$$= 1.46224$$

$$X_2 = x_0 + 2.75(0.1524 - 0) = 0.4191$$

_/_/_

Q Data has been collected on service times at a single bill payment window in a bank. The service time data are summarized into ~~inter~~ intervals as follows

	Intervals	Freq	Prob	Cum. Prob	Slope
1	30 - 60	10	0.1	0.1	300
2	60 - 90	25	0.25	0.35	120
3	90 - 120	35	0.35	0.70	85.71
4	120 - 180	20	0.2	0.9	300
5	180 - 300	10	0.1	1.0	1200

$R_1 = 0.3456$, $R_2 = 0.7651$

$x_i = x_{i-1} + a_i (R - c_{i-1})$

$R_1 = 0.3456$, $i = 2$

$x_i = 89.472$

$R_2 = 0.7651$, $i = 4$

$x_i = 120 + 300(0.7651 - 0.70) = 139.53$

Discrete Distribution

- All discrete distributions can be generated via inverse-transform technique
- Method: numerically, table-lookup procedure, algebraically — or a formula.

- Examples → Poisson, Geometric

_/_/_

* Empirical Discrete Distribution

① Generate uniform random no.s, $R_1, R_2 \ldots$

$R \sim U[0,1]$

② Return desired random variate $x = i$ if it satisfies $\sum_{j=0}^{i-1} p(i) < R \leq \sum_{j=0}^{i} p(i)$ and $F(x-1) < R \leq F(x)$

Q At the end of the week, data is generated on the no. of items bought by customers from a supermarket. The no. of items bought is either 1, 2, 3, 4, 5 with relative freq. 0.10, 0.40, 0.05, 0.15, 0.30 respectively. Set up the table for generating the no. of items using table lookup method and generate 2 values of no of items bought using uniform random no.s $R_1 = 0.8$, $R_2 = 0.35$

No. of Items	Freq	Cum Freq
1	0.10	0.10
2	0.40	0.50
3	0.05	0.55
4	0.15	0.70
5	0.30	1.00

$R_1 = 0.80$

$0.7 < R_1 < 1.0$

$i = 5$

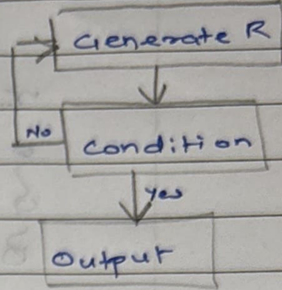
$R_2 = 0.35$

$0.10 < R_2 < 0.50$

$i = 2$

Acceptane- Rejection Technique

- useful particularly when inverse cdf does not exist in closed form, aka thinning



- Step 1: Generate $R \sim U[0,1]$
  Step 2a: If $R >= 1/4$, accept $X = R$
  Step 2b: If $R < 1/4$, reject → Step 1

Efficiency: Depends heavily on ability to minimize no. of rejections

Poisson Distribution

- Set $x = 0$, $P = 1$
- Generate random no. $R_{x+1} \sim U[0,1]$ and replace P by $P \times (R_{x+1})$
- If $P < e^{-\lambda}$ then accept $X = x$
- Else reject x and increment x by 1 and return to step 2

Q The no. of customers arriving at a bank b/w 10 am and 1am is poisson distributed with mean = 3. Generate poisson variates using random no.s, $R_1 = 0.5294$, $R_2 = 0.3578$, $R_3 = 0.1574$

→ $x = 0$, $P = 1$, $e^{-\lambda} = e^{-3} = 0.0497$

$R_1 = 0.5294$,

$P = P \times R_1 = 1 \times 0.5294 = 0.5294$

$P < e^{-\lambda}$ → false → reject

$R_2 = 0.3578$

$P = 0.3578 \times 0.5294$

$= 0.18941$ → $P < e^{-\lambda}$ → false

$R_3 = 0.1574$

$P = 0.18941 \times 0.1574 = 0.029814$

P → accept

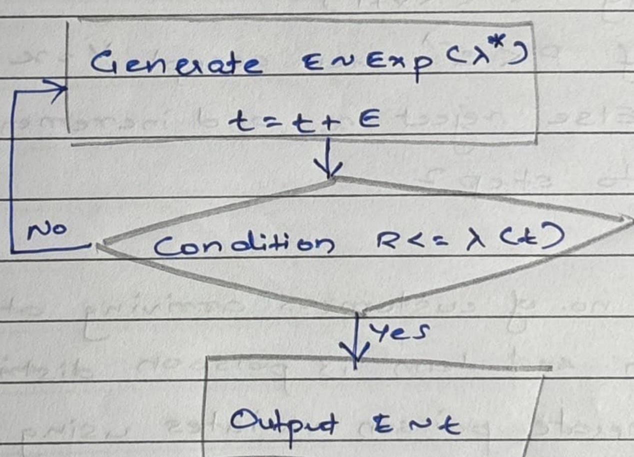
x	P	R_{x+1}	Accept $e^{-\lambda} > P$	Res
0	1	0.5294	false	
1	0.5294	0.3578	false	
2	0.1894	0.1574	false	
3	0.02984	-	true	N=3

Q mean = 0.2, R_1 = 0.4357, R_2 = 0.4146, R_3 = 0.8353, 0.9952,
R_5 = 0.8004

x	R_{x+1}	P	Accept / Rej	Result
0	0.4357	0.4357	acc	N=0
0	0.4146	0.4146	acc	N=0
0	0.8353	0.8353	rej	
2	0.9952	0.8313	rej	
	0.8004	0.6654	acc	N=2

NSPP → Non Stationary Poisson Process

↳ Generate a stationary Poisson arrival process at the fastest rate, $\lambda^* = \max \lambda(t)$



Q Generate random variates for a non-stationary poisson process using the data given below

t	0	60	120	180	240	300	360	420	480
mean time	15	12	7	5	8	10	15	20	20
arrival	1/15	1/12	1/7	1/5	1/8	1/10	1/15	1/20	1/20

Random no.s given: 0.2130, 0.8830, 0.5530, 0.0240, 0.0001, 0.14443

$\rightarrow \lambda^* = \max \lambda(t) = \frac{1}{5}, \; t = 0, \; i = 1$

for random no., $R_1 = 0.2130$

$E = -5 \ln(0.213) = 13.13$

$t = t + E = 13.13$

$R = 0.8830$

$\lambda(13.13) / \lambda^* = 1/15 \,/\, 1/5 = 1/3$

$R > 1/3 \rightarrow$ do not generate arrival

for random no. $R = 0.5530$

$E = -5 \ln(0.553) = 2.96$

$t = 13.13 + 2.96 = 16.09$

Generate $R = 0.0240$

$\lambda(16.09) / \lambda^* = (1/15) / (1/5) = 1/3$

$R < 1/3, \; T_1 = t = 16.09$

$i = i + 1 = 1 + 1 = 2$

for random no. $R = 0.0001$

$E = -5 \ln(0.001) = 46.05$

$t = 16.09 + 46.05 = 62.14$

Generate $R = 0.1443 < 1/12 \times 1/15$

$T_1 = 62.14$